ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

DIGITAL ELECTRONICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 1st year, semester 2 final examination: number systems and codes, logic gates and Boolean algebra, Karnaugh maps and design, combinational building blocks, sequential logic, logic families and interfacing.

Written by Academic Central for practice, following first-year Digital Electronics (digital systems) syllabuses at South African universities (BEng Electrical, Electronic and Mechatronic Engineering). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Show ALL your working for conversions and simplifications.
3. Draw logic circuits neatly, using standard (distinctive-shape) gate symbols.
4. Non-programmable calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: NUMBER SYSTEMS AND CODES

1.1 Convert 156_{10} to binary and to hexadecimal. (3)

1.2 Convert $1011\ 0110_{2}$ to decimal. (2)

1.3 Convert $3AF_{16}$ to decimal. (2)

1.4 Write −45 as an 8-bit two's complement number. (3)

1.5 Add the binary numbers 0110 1011 and 0011 0101, and check your answer in decimal. (2)

1.6 Write 47_{10} in BCD (8421 code). (1)

1.7 Why is Gray code used in rotary shaft encoders? Convert the binary number 0111 to Gray code. (2)

[15]

QUESTION 2: LOGIC GATES AND BOOLEAN ALGEBRA

2.1 Draw the truth tables of a 2-input NAND gate, NOR gate and XOR gate. (3)

2.2 Why is the NAND gate called a universal gate? Show how a NOT gate and an AND gate can be made from NAND gates only. (3)

2.3 Use Boolean algebra to simplify Y = AB + AB′ + A′B. Name the law used in each step. (4)

2.4 Use a truth table to prove De Morgan's theorem (A + B)′ = A′ · B′. (3)

Answer QUESTION 2.5 using the logic circuit in FIGURE 1.

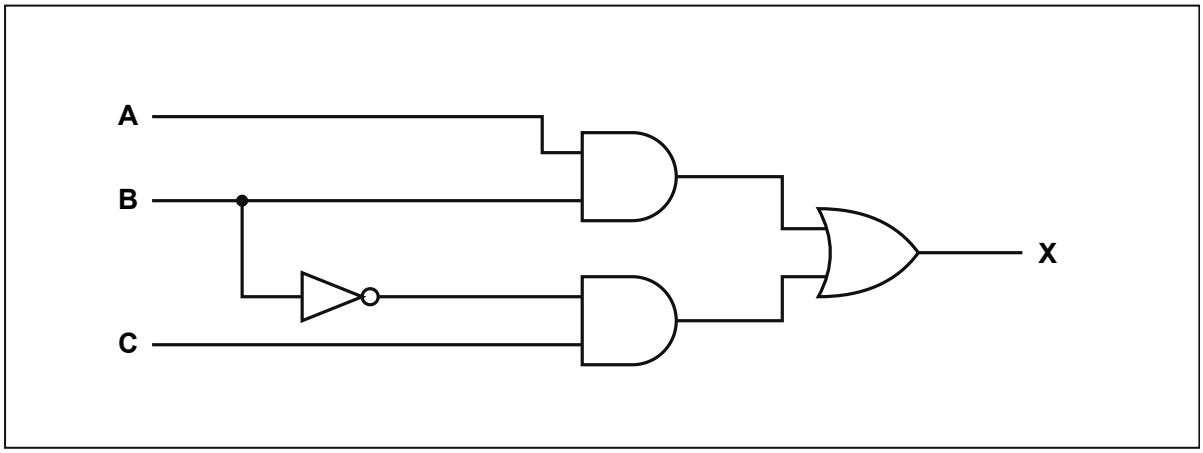


FIGURE 1: Logic circuit

2.5 Answer the following:

2.5.1 Write down the Boolean expression for the output X. (2)

2.5.2 Draw the truth table of the circuit. (4)

2.5.3 What is the output when A = 1, B = 0 and C = 1? (1)

[20]

QUESTION 3: KARNAUGH MAPS AND DESIGN

3.1 Use a Karnaugh map to simplify F(A, B, C, D) = Σm(0, 2, 5, 7, 8, 10, 13, 15). Show your groups clearly. (6)

3.2 A robot uses three obstacle sensors A, B and C. To avoid false alarms, the warning output Y must be 1 only when at least TWO of the three sensors give a 1.

3.2.1 Draw the truth table. (3)

3.2.2 Write down the unsimplified sum-of-products expression. (2)

3.2.3 Simplify the expression. (3)

3.2.4 Draw the simplified logic circuit using AND and OR gates. (3)

3.2.5 Explain how the circuit can be built using NAND gates only. (3)

[20]

QUESTION 4: COMBINATIONAL BUILDING BLOCKS

4.1 Draw the truth table of a half adder and write down the expressions for its sum (S) and carry (C) outputs. (4)

4.2 Explain how a full adder can be made from two half adders and an OR gate. Give the outputs of a full adder when A = 1, B = 1 and C_{in} = 1. (3)

4.3 A 4-to-1 multiplexer has data inputs I_0 to I_3 and select inputs S_1 and S_0.

4.3.1 Write down the Boolean expression for its output Y. (3)

4.3.2 Which input is connected to the output when S_1S_0 = 10? (1)

4.4 Give ONE use of a 2-to-4 decoder in a microcontroller system. (2)

4.5 Which segments (a to g) of a seven-segment display light up to show the digit 7? (2)

[15]

QUESTION 5: SEQUENTIAL LOGIC

5.1 Explain the difference between combinational and sequential logic circuits. (2)

5.2 Draw the truth table of an SR latch made from NOR gates and explain which input combination is not allowed and why. (4)

5.3 A positive-edge-triggered D flip-flop starts with Q = 0. The values of D at five rising clock edges are 1, 0, 1, 1, 0. Write down Q after each edge, and explain what "edge-triggered" means. (3)

5.4 Draw the truth table of a JK flip-flop and explain the toggle mode. (4)

5.5 A 3-bit binary ripple counter is built from flip-flops.

5.5.1 How many different states does it have, and what is its highest count? (2)

5.5.2 How can it be changed into a MOD-6 counter? (1)

5.6 A 1 MHz clock drives a 4-bit binary counter. Calculate the frequency of the signal at the most significant bit output. (2)

5.7 A 4-bit serial-in, parallel-out shift register starts cleared. Data enters at Q_0 and shifts towards Q_3 on each clock pulse. The bits 1, 0, 1, 1 are shifted in, in that order. Write down $Q_3Q_2Q_1Q_0$ after the fourth clock pulse. (2)

[20]

QUESTION 6: LOGIC FAMILIES AND INTERFACING

6.1 Compare TTL and CMOS logic families under the headings supply voltage, power use, and sensitivity to static electricity. (4)

6.2 Define fan-out. (2)

6.3 A 5 V microcontroller pin drives an LED that needs 10 mA and has a forward voltage of 2 V. Calculate the series resistor needed and choose a standard value. (2)

6.4 Why is a pull-up resistor connected to a microcontroller input that reads a push button? (2)

[10]

TOTAL: 100

END OF QUESTION PAPER